

MEDICAL HANDBOOK - CLINICAL REFERENCE GUIDE
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Classification: FOR MEDICAL PROFESSIONALS ONLY

SECTION 1: COMMON COLD & UPPER RESPIRATORY INFECTION MANAGEMENT

1.1 OVERVIEW OF COMMON COLD TREATMENT

The common cold is a viral upper respiratory tract infection primarily caused by rhinoviruses. Symptoms include rhinorrhea, nasal congestion, sneezing, sore throat, cough, headache, and low-grade fever. Treatment is symptomatic.

1.2 FIRST-LINE SYMPTOMATIC TREATMENTS

Medication Class	Examples	Target Symptoms
Analgesics/Antipyretics	Acetaminophen 500mg Ibuprofen 200-400mg	Fever, headache, body ache Fever, inflammation, pain
Oral Decongestants	Pseudoephedrine 60mg	Nasal congestion
Nasal Decongestants	Oxymetazoline 0.05%	Nasal congestion (topical)
Antihistamines (1st gen)	Diphenhydramine 25mg	Sneezing, rhinorrhea
Antihistamines (2nd gen)	Loratadine 10mg Cetirizine 10mg Fexofenadine 180mg	Sneezing, rhinorrhea (non-drowsy) Sneezing, rhinorrhea Sneezing, rhinorrhea
Cough Suppressants	Dextromethorphan 15mg	Dry cough
Expectorants	Guaifenesin 200-400mg	Productive cough, chest congestion
Mucolytics	Acetylcysteine 600mg	Thick mucus, productive cough
Throat Lozenges	Benzocaine lozenges	Sore throat
Combination Products	Various	Multi-symptom relief

SECTION 2: DRUG INTERACTION PROFILES - COLD MEDICATIONS WITH CHRONIC CONDITIONS

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2.1 HYPERTENSION - SAFE COLD MEDICATIONS

RISK ASSESSMENT: HIGH - Many cold medications can elevate blood pressure

SAFE OPTIONS (First Choice):

- Acetaminophen (Paracetamol): NO effect on blood pressure. Safe for all HTN patients.
Dosage: 500-1000mg q6h, max 3g/day (2g/day if hepatic impairment)
- Guaifenesin: NO cardiovascular effects. Safe expectorant.
Dosage: 200-400mg q4-6h, max 2400mg/day
- Dextromethorphan: NO significant BP effects at therapeutic doses.
Dosage: 10-20mg q4h, max 120mg/day
- Nasal Saline Spray: Completely safe, no drug interactions.
- Oxymetazoline (topical): Minimal systemic absorption. Use max 3 days.
Dosage: 2-3 sprays each nostril q12h

USE WITH CAUTION:

- Loratadine/Cetirizine/Fexofenadine: Generally safe. No significant BP effects.
- Benzonatate: Safe for cough. No cardiovascular effects.
Dosage: 100-200mg TID, max 600mg/day

CONTRAINDICATED (DO NOT USE):

- Pseudoephedrine: SIGNIFICANT BP ELEVATION. Alpha-adrenergic agonist.
Risk: Can increase SBP by 5-15 mmHg. May reduce antihypertensive efficacy.
AVOID in: Uncontrolled HTN, Stage 2 HTN, HTN with CVD
- Phenylephrine (oral): Moderate BP elevation risk.
- Ephedrine: Contraindicated.
- High-dose NSAIDs: Sodium retention, reduced antihypertensive effect.

INTERACTION WITH ACE INHIBITORS (Lisinopril, Enalapril, Ramipril):

- Pseudoephedrine: Reduces antihypertensive effect. AVOID.
- NSAIDs: Can reduce ACEi efficacy and increase renal risk. Use acetaminophen instead.
- Dextromethorphan: Generally safe. No CYP interaction with ACEi.
- Guaifenesin: Safe. No interaction.

INTERACTION WITH BETA-BLOCKERS (Metoprolol, Atenolol, Carvedilol):

- Pseudoephedrine: Opposing cardiac effects. Can cause tachycardia and HTN. AVOID.
- Dextromethorphan: Safe. No interaction.
- NSAIDs: Can reduce beta-blocker efficacy. Monitor BP.

INTERACTION WITH CALCIUM CHANNEL BLOCKERS (Amlodipine, Nifedipine):

- Pseudoephedrine: Counteracts vasodilation. AVOID.
- Other cold medications generally safe.

INTERACTION WITH DIURETICS (Furosemide, HCTZ, Chlorthalidone):

- NSAIDs: Can reduce diuretic efficacy and increase nephrotoxicity risk. AVOID.
- Acetaminophen: Safe. Preferred analgesic.
- Pseudoephedrine: Use with caution. Monitor BP.

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2.2 DIABETES MELLITUS - SAFE COLD MEDICATIONS

RISK ASSESSMENT: MODERATE - Focus on blood glucose monitoring during illness

SAFE OPTIONS (First Choice):

- Acetaminophen: NO effect on blood glucose. Safe.
- Guaifenesin: NO effect on glucose. Safe.
- Dextromethorphan: NO effect on glucose. Safe.
- Loratadine/Cetirizine/Fexofenadine: No glucose effects. Safe.
- Nasal saline spray: Safe.
- Oxymetazoline nasal: Safe. No systemic effects on glucose.

USE WITH CAUTION:

- Pseudoephedrine: May cause mild hyperglycemia. Monitor glucose.
- Phenylephrine: Similar caution as pseudoephedrine.

CONTRAINDICATED/AVOID:

- Oral corticosteroids (Prednisone, Dexamethasone): SIGNIFICANT hyperglycemia.
Can increase blood glucose by 100-200 mg/dL.
ONLY use if absolutely necessary with insulin adjustment.
- Cough syrups with sugar/alcohol: Check labels. Use sugar-free formulations.
- High-dose Aspirin: Can potentiate sulfonylurea effects.

INTERACTION WITH METFORMIN:

- No direct drug interactions with common cold medications.
- Alcohol-containing cough syrups: AVOID (increased lactic acidosis risk).
- Dehydration risk with fever: Monitor renal function. Consider holding metformin if severe vomiting/diarrhea.

INTERACTION WITH INSULIN:

- Prednisone: Will increase insulin requirements by 30-50%.
- Infection/fever itself: May increase insulin requirements. Monitor closely.
- Acetaminophen: No interaction with insulin.

INTERACTION WITH SULFONYLUREAS (Glipizide, Glyburide, Glimepiride):

- Aspirin (high dose): Can displace sulfonylureas from protein binding → hypoglycemia.
- NSAIDs: Similar risk. Use acetaminophen instead.
- Alcohol: AVOID (hypoglycemia risk).

INTERACTION WITH DPP-4 INHIBITORS (Sitagliptin, Saxagliptin):

- No significant drug interactions with cold medications.
- Safe to use all first-line cold treatments.

INTERACTION WITH SGLT2 INHIBITORS (Empagliflozin, Dapagliflozin):

- Dehydration risk with fever: Increased risk of DKA. Ensure adequate hydration.
- NSAIDs: Increased renal risk. Avoid.

INTERACTION WITH GLP-1 AGONISTS (Semaglutide, Liraglutide):

- No direct drug interactions with cold medications.
- Nausea from cold + GLP-1: May compound GI side effects.

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2.3 CHRONIC KIDNEY DISEASE (CKD) - SAFE COLD MEDICATIONS

RISK ASSESSMENT: HIGH - Many medications require dose adjustment or are contraindicated

SAFE OPTIONS:

- Acetaminophen: Analgesic of choice. Safe in all CKD stages.
Dosage: 500-650mg q6-8h, max 3g/day (Stage 1-2), max 2g/day (Stage 3-4)
- Guaifenesin: Safe. Excreted renally but no nephrotoxicity.

- Dextromethorphan: Safe. Hepatic metabolism.
- Loratadine: Safe. No renal adjustment needed.
- Cetirizine: Safe but may accumulate. Reduce to 5mg in Stage 4-5 CKD.
- Oxymetazoline nasal: Safe. No systemic accumulation.
- Saline nasal spray: Safe.

USE WITH CAUTION:

- Fexofenadine: Reduce dose in CKD Stage 4-5.
- Benzonatate: Limited data. Use with caution.

CONTRAINDICATED:

- NSAIDs (Ibuprofen, Naproxen, Diclofenac, Ketorolac):
ABSOLUTELY CONTRAINDICATED in CKD Stage 3-5.
Risk: Acute kidney injury, further GFR decline, hyperkalemia, sodium retention.
Even short courses (3-5 days) can cause significant renal injury in CKD patients.
- Pseudoephedrine: Can increase blood pressure and reduce renal blood flow.
Avoid in CKD with hypertension.
- High-dose Aspirin: Avoid in CKD Stage 4-5.

INTERACTION WITH DIURETICS IN CKD (Furosemide, Bumetanide):

- NSAIDs: TRIPLE WHAMMY EFFECT (NSAID + Diuretic + ACEi/ARB) → High AKI risk.
AVOID COMPLETELY.
- Acetaminophen: Safe with diuretics.

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2.4 ATRIAL FIBRILLATION & ANTICOAGULANT USE - SAFE COLD MEDICATIONS

RISK ASSESSMENT: HIGH - Drug interactions with anticoagulants and cardiac effects

SAFE OPTIONS:

- Acetaminophen: First-line analgesic. Safe with anticoagulants at recommended doses.
CRITICAL: Limit to 2g/day maximum. Chronic high-dose (>2g/day) can increase INR.
- Guaifenesin: Safe. No interaction with anticoagulants.
- Loratadine/Cetirizine: Safe. No interaction with warfarin or DOACs.
- Nasal saline: Safe.
- Oxymetazoline nasal: Safe for short-term use (max 3 days).

USE WITH CAUTION:

- Dextromethorphan: Generally safe but use caution with apixaban/rivaroxaban (theoretical CYP competition, not clinically significant).

CONTRAINDICATED:

- NSAIDs: ABSOLUTELY CONTRAINDICATED with anticoagulants.
Risk: Major bleeding (GI, intracranial). Synergistic anticoagulant effect.
Even single doses significantly increase bleeding risk.
- Aspirin: Generally contraindicated unless specifically prescribed.
Added bleeding risk without additional cardiovascular benefit.

INTERACTION WITH WARFARIN:

- Acetaminophen: Monitor INR if using >2g/day for >3 days.
- Dextromethorphan: No clinically significant interaction.
- Guaifenesin: Safe. No interaction.
- Cranberry products (some cold remedies): Can potentiate warfarin. Avoid.

INTERACTION WITH DOACs (Apixaban, Rivaroxaban, Edoxaban, Dabigatran):

- Acetaminophen: Safe at standard doses.
- P-glycoprotein interactions: Dextromethorphan and antihistamines generally safe.
- NSAIDs: ABSOLUTELY CONTRAINDICATED.

INTERACTION WITH ANTIPLATELETS (Clopidogrel, Prasugrel, Ticagrelor):

- Acetaminophen: Safe. First-line analgesic.
- NSAIDs: Contraindicated. Increased bleeding risk.

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2.5 ASTHMA & COPD - SAFE COLD MEDICATIONS

RISK ASSESSMENT: MODERATE - Respiratory medications and cold treatments interactions

SAFE OPTIONS:

- Acetaminophen: Safe. No respiratory effects.
- Guaifenesin: Safe and beneficial. Helps clear mucus.
- Acetylcysteine: Safe and beneficial. Mucolytic.
- Dextromethorphan: Safe for cough. No bronchial effects.
- Loratadine/Cetirizine/Fexofenadine: Safe.

- Oxymetazoline nasal: Safe. No bronchial effects.

USE WITH CAUTION:

- Pseudoephedrine: Generally safe in asthma but use caution.
Theoretical bronchodilation (beta-agonist effect).

CONTRAINDICATED/AVOID:

- Beta-blocker cold preparations (rare): AVOID in asthma.
- Aspirin: Caution in aspirin-sensitive asthma (Samter's Triad).
- NSAIDs: Caution in aspirin-sensitive patients.

INTERACTION WITH BETA-AGONISTS (Albuterol, Salmeterol, Formoterol):

- Pseudoephedrine: Additive cardiovascular stimulation. Monitor heart rate.
- Dextromethorphan: Safe. No interaction.
- Guaifenesin: Safe and complementary.

INTERACTION WITH INHALED CORTICOSTEROIDS (Fluticasone, Budesonide):

- No significant drug interactions with any cold medications.
- Continue ICS during cold illness.

INTERACTION WITH TIOTROPIUM/ANTICHOLINERGICS:

- First-generation antihistamines (Diphenhydramine): Additive anticholinergic effects.
Use second-generation antihistamines instead (Loratadine, Cetirizine).
- Other cold medications generally safe.

INTERACTION WITH THEOPHYLLINE:

- Pseudoephedrine: Additive CNS stimulation. AVOID.
- Dextromethorphan: Safe.
- Macrolide antibiotics (if prescribed): Can increase theophylline levels.

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2.6 DEPRESSION & ANXIETY (SSRI/SNRI USE) - SAFE COLD MEDICATIONS

RISK ASSESSMENT: HIGH - Serotonin syndrome risk with certain cold medications

SAFE OPTIONS:

- Acetaminophen: Safe. No interaction with antidepressants.
- Guaifenesin: Safe. No serotonin effects.

- Loratadine/Cetirizine/Fexofenadine: Safe. No serotonin interaction.
- Nasal decongestants (topical): Safe. Oxymetazoline.
- Nasal saline: Safe.
- Benzonatate: Safe for cough.

SERIOUS INTERACTION - CONTRAINDICATED:

- DEXTROMETHORPHAN + SSRIs (Sertraline, Fluoxetine, Paroxetine, Citalopram, Escitalopram)

= SEROTONIN SYNDROME RISK

Mechanism: Dextromethorphan is a serotonin reuptake inhibitor.

Combined with SSRIs → excessive serotonin activity.

Symptoms: Agitation, confusion, hyperthermia, tachycardia, myoclonus, rigidity.

ONSET: Can occur within hours of combination.

ABSOLUTE CONTRAINDICATION - DO NOT COMBINE.

- DEXTROMETHORPHAN + SNRIs (Venlafaxine, Duloxetine)

= Same serotonin syndrome risk. CONTRAINDICATED.

- DEXTROMETHORPHAN + MAOIs

= EXTREME RISK. Can be fatal. ABSOLUTELY CONTRAINDICATED.

USE WITH CAUTION:

- Pseudoephedrine + SSRIs/SNRIs: Possible increased sympathomimetic effects.

Monitor for anxiety, tachycardia, hypertension.

- Phenylephrine + SSRIs: Similar caution as pseudoephedrine.

INTERACTION WITH SPECIFIC ANTIDEPRESSANTS:

- Sertraline (Zoloft): AVOID dextromethorphan. All other cold meds safe.

- Fluoxetine (Prozac): AVOID dextromethorphan. Long half-life - risk persists for weeks.

- Paroxetine (Paxil): AVOID dextromethorphan.

- Escitalopram (Lexapro): AVOID dextromethorphan.

- Bupropion (Wellbutrin): Dextromethorphan safe (no serotonin effect).

But bupropion inhibits CYP2D6 → increased dextromethorphan levels.

Use half-dose dextromethorphan or avoid.

- Mirtazapine (Remeron): Dextromethorphan caution. Low risk but possible.

- Trazodone: Dextromethorphan caution. Additive serotonergic effects possible.

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2.7 GERD - SAFE COLD MEDICATIONS

RISK ASSESSMENT: LOW - Few direct drug interactions

SAFE OPTIONS:

- All first-line cold medications generally safe with GERD.
- Acetaminophen: Safe.
- Guaifenesin: Safe.
- Antihistamines: Safe.

NOTE: Cough from cold + GERD can be additive. Treat both conditions.
Avoid alcohol-containing cough syrups (worsens GERD).

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2.8 GOUT - SAFE COLD MEDICATIONS

RISK ASSESSMENT: MODERATE - NSAID and diuretic interactions

SAFE OPTIONS:

- Acetaminophen: Safe. First-line analgesic.
- Guaifenesin: Safe.
- Dextromethorphan: Safe.
- Antihistamines: Safe.
- Nasal preparations: Safe.

CONTRAINDICATED/AVOID:

- Aspirin (low-dose for cardioprotection is usually continued but higher doses avoided):
Low-dose (81mg) can increase uric acid. Discuss risk/benefit.
High-dose aspirin: AVOID. Increases uric acid levels.
- Diuretic-containing cold preparations (rare): Can increase uric acid.

INTERACTION WITH ALLOPURINOL/FEBUXOSTAT:

- No significant interactions with standard cold medications.
- Acetaminophen safe. No interaction with xanthine oxidase inhibitors.

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2.9 HYPOTHYROIDISM - SAFE COLD MEDICATIONS

RISK ASSESSMENT: LOW - Few interactions with levothyroxine

SAFE OPTIONS:

- All standard cold medications generally safe with levothyroxine.
- Acetaminophen: Safe.
- Guaifenesin: Safe.
- Dextromethorphan: Safe.
- Antihistamines: Safe.

USE WITH CAUTION:

- Pseudoephedrine: Can cause tachycardia. Caution in patients with cardiac symptoms.
- Phenylephrine: Similar caution.

INTERACTION WITH LEVOTHYROXINE:

- No clinically significant interactions with cold medications.
- Maintain levothyroxine schedule. Take on empty stomach as usual.
- Iron-containing preparations: Separate from levothyroxine by 4 hours.

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2.10 MIGRAINE - SAFE COLD MEDICATIONS

RISK ASSESSMENT: MODERATE - Vasoconstrictor interactions with triptans

SAFE OPTIONS:

- Acetaminophen: Safe. First-line analgesic.
- Guaifenesin: Safe.
- Dextromethorphan: Safe.
- Loratadine/Cetirizine/Fexofenadine: Safe.
- Nasal saline: Safe.

CONTRAINDICATED:

- Pseudoephedrine + Triptans (Sumatriptan, Rizatriptan, etc.):

ADDITIVE VASOCONSTRICTION RISK.

Both are vasoconstrictors → increased risk of vasospasm, hypertension, and potentially coronary vasospasm.

AVOID COMBINATION within 24 hours.

- Phenylephrine + Triptans:
Similar vasoconstriction risk. Avoid.

INTERACTION WITH SUMATRIPTAN:

- Pseudoephedrine: CONTRAINDICATED within 24 hours.
- Phenylephrine: CONTRAINDICATED within 24 hours.
- Oxymetazoline nasal: LOW systemic absorption. Probably safe but caution.
- All other cold meds: Safe.

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2.11 ANXIETY DISORDERS - SAFE COLD MEDICATIONS

RISK ASSESSMENT: MODERATE - CNS stimulant effects of decongestants

SAFE OPTIONS:

- Acetaminophen: Safe.
- Guaifenesin: Safe.
- Second-generation antihistamines (Loratadine, Cetirizine): Safe. Non-sedating.
- Nasal preparations: Safe.

USE WITH CAUTION:

- Pseudoephedrine: Can cause or worsen anxiety, nervousness, insomnia.
Use with caution in anxiety patients.
- Phenylephrine: Similar anxiogenic potential, though less than pseudoephedrine.
- First-generation antihistamines (Diphenhydramine): Can cause drowsiness.
May interact with anxiolytics additively.

INTERACTION WITH BENZODIAZEPINES (Alprazolam, Lorazepam, etc.):

- Diphenhydramine: Additive sedation. Avoid or use with caution.
- Second-generation antihistamines: Safe. No additive sedation.
- Dextromethorphan: Safe. No interaction.
- Pseudoephedrine: May counteract sedation. Complex interaction.

INTERACTION WITH BUSPIRONE:

- Dextromethorphan: Safe. Buspirone is a 5-HT_{1A} agonist, not significant SRI.
Low serotonin syndrome risk, but monitor.

- Other cold meds generally safe.

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SECTION 3: COMPREHENSIVE DRUG INTERACTION TABLES

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3.1 DECONGESTANTS - DRUG INTERACTION MATRIX

Drug Class/Medication	Pseudoephedrine	Phenylephrine(Oral)	Oxymetazoline(Nasal)
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ACE Inhibitors	HIGH RISK (↓BP ctrl)	MODERATE RISK	LOW RISK (safe ≤3 days)
ARBs	HIGH RISK (↓BP ctrl)	MODERATE RISK	LOW RISK (safe ≤3 days)
Beta-Blockers	HIGH RISK (opposing)	MODERATE RISK	LOW RISK
CCBs	HIGH RISK (opposing)	MODERATE RISK	LOW RISK
Diuretics	MODERATE RISK	MODERATE RISK	LOW RISK
Triptans (Migraine)	CONTRAINDICATED	CONTRAINDICATED	LOW RISK (caution)
MAOIs	CONTRAINDICATED	CONTRAINDICATED	LOW RISK
SSRIs	MODERATE RISK	MODERATE RISK	LOW RISK
Antidiabetics	LOW (monitor glucose)	LOW (monitor glucose)	LOW RISK
Warfarin	LOW RISK	LOW RISK	LOW RISK
DOACs	LOW RISK	LOW RISK	LOW RISK
Levothyroxine	MODERATE (tachy.)	MODERATE (tachy.)	LOW RISK

3.2 DEXTROMETHORPHAN - DRUG INTERACTION MATRIX

Drug Class/Medication	Risk Level	Interaction
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SSRIs (all)	CONTRAINDICATED	Serotonin Syndrome
SNRIs (Venlafaxine, etc.)	CONTRAINDICATED	Serotonin Syndrome
MAOIs	CONTRAINDICATED	Serotonin Syndrome (fatal)
Bupropion	CAUTION	↑ Dextromethorphan levels
Tramadol	CONTRAINDICATED	Serotonin Syndrome
Linezolid	CONTRAINDICATED	Serotonin Syndrome
Mirtazapine	CAUTION	Possible serotonergic

Trazodone	CAUTION	Additive serotonergic
ACE Inhibitors	SAFE	No interaction
ARBs	SAFE	No interaction
Beta-Blockers	SAFE	No interaction
CCBs	SAFE	No interaction
Warfarin	SAFE	No interaction
DOACs	SAFE (caution)	Theoretical CYP
Insulin	SAFE	No interaction
Metformin	SAFE	No interaction
Levothyroxine	SAFE	No interaction
Antihistamines	SAFE	No interaction

3.3 NSAIDs - DRUG INTERACTION MATRIX

Drug Class/Medication	Risk Level	Interaction
Anticoagulants (all)	CONTRAINDICATED	Major Bleeding Risk
Antiplatelets	CONTRAINDICATED	Major Bleeding Risk
CKD Stage 3-5	CONTRAINDICATED	AKI, Nephrotoxicity
ACE Inhibitors	HIGH RISK	↓Antihypertensive, ↑Renal risk
ARBs	HIGH RISK	↓Antihypertensive, ↑Renal risk
Diuretics	HIGH RISK	↓Diuretic effect, ↑Nephrotoxicity
Lithium	HIGH RISK	↑Lithium levels (toxicity)
Methotrexate	HIGH RISK	↑Methotrexate toxicity
SSRIs	HIGH RISK	↑Bleeding risk
Corticosteroids	HIGH RISK	↑GI bleeding risk
Gout (Allopurinol)	SAFE (caution)	Aspirin avoided, others cautious
Levothyroxine	SAFE	No interaction
Metformin	SAFE	No interaction
Antihistamines	SAFE	No interaction

SECTION 4: CLINICAL DECISION SUPPORT - COLD TREATMENT ALGORITHMS

4.1 COLD TREATMENT PROTOCOL FOR HYPERTENSION + DIABETES PATIENT

Patient Profile: Hypertension on ACEi/ARB + Type 2 Diabetes on Metformin

Presenting: Sneezing, rhinorrhea, mild cough, no fever

STEP 1: Assess symptom severity

STEP 2: Select medications based on safety profile:

RECOMMENDED REGIMEN:

Symptom	Medication	Dosage	Notes
Sneezing/Runny Nose	Loratadine	10mg once daily	First-line. Non-drowsy.
Nasal Congestion	Oxymetazoline nasal OR Saline nasal spray	2 sprays each nostril As needed	Max 3 days only. Unlimited use.
Cough (dry)	Dextromethorphan	15mg q6-8h	Max 60mg/day.
Cough (productive)	Guaifenesin	200mg q4h	Max 1200mg/day.
Fever/Body Ache	Acetaminophen	500mg q6h	Max 2g/day (if liver OK).
Sore Throat	Warm saline gargle	q4h	Safe. No drug interactions.

MEDICATIONS TO AVOID:

- ✗ Pseudoephedrine - Increases BP, reduces antihypertensive efficacy
- ✗ Phenylephrine (oral) - Same as above, milder but still risk
- ✗ NSAIDs (Ibuprofen, Naproxen) - Reduces ACEi efficacy, renal risk with diabetes
- ✗ Combination cold products containing above ingredients
- ✗ High-sugar cough syrups (if diabetes not well controlled)

MONITORING:

- BP: Check daily during cold illness
- Blood glucose: Check q4-6h during illness (illness can increase glucose)
- Hydration: Encourage fluids, especially if fever present

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4.2 COLD TREATMENT PROTOCOL FOR PATIENT ON ANTICOAGULANTS

Patient Profile: Atrial Fibrillation on Apixaban + Hypertension on Metoprolol + Gout
Presenting: Fever, body ache, sneezing, mild cough

RECOMMENDED REGIMEN:

Symptom	Medication	Dosage	Notes
Fever/Body Ache	Acetaminophen	500mg q6h	MAX 2g/day. Do not exceed.
Sneezing	Loratadine	10mg daily	Safe with anticoagulants.
Cough	Guaifenesin	200mg q4h	Safe.
Nasal Congestion	Saline nasal spray	As needed	Safe.

MEDICATIONS TO AVOID:

✗ NSAIDs (Ibuprofen, Naproxen, Aspirin) - ABSOLUTELY CONTRAINDICATED with Apixaban

Risk: Major bleeding (GI bleed, ICH). Even single dose significantly increases risk.

✗ Pseudoephedrine - Opposes metoprolol, may increase BP

✗ High-dose Acetaminophen (>2g/day) - Can increase bleeding risk with anticoagulants

CRITICAL WARNINGS:

- Any sign of bleeding (bruising, dark stools, nosebleeds) → Immediate medical attention
- Do not exceed 2g Acetaminophen per day
- Avoid all NSAID-containing products including topical

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4.3 COLD TREATMENT PROTOCOL FOR PATIENT ON ANTIDEPRESSANTS (SSRI)

Patient Profile: Depression on Sertraline + Hypertension + GERD

Presenting: Cough (dry), runny nose, sneezing

RECOMMENDED REGIMEN:

Symptom	Medication	Dosage	Notes
Cough (dry)	Guaifenesin	200mg q4h	Safe. No serotonin effect.
	OR Benzonatate	100mg TID	Safe. Alternative for cough.
Runny Nose/Sneezing	Loratadine	10mg daily	Safe. Non-sedating.
	OR Cetirizine	10mg daily	Safe.
Nasal Congestion	Oxymetazoline nasal	2 sprays q12h	Max 3 days only.
Fever/Body Ache	Acetaminophen	500mg q6h	Safe.

MEDICATIONS TO AVOID:

X DEXTROMETHORPHAN - SEROTONIN SYNDROME RISK. ABSOLUTELY CONTRAINDICATED.

Mechanism: DXM is SRI + Sertraline is SRI = excessive serotonin.

Can occur within hours. Can be life-threatening.

X Pseudoephedrine - Can increase anxiety, nervousness, hypertension

X Phenylephrine - Similar concerns, milder

X Combination cold products (many contain dextromethorphan) - CHECK LABELS

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4.4 COLD TREATMENT PROTOCOL FOR CKD PATIENT

Patient Profile: CKD Stage 2-3 + Diabetes + Hyperlipidemia

Presenting: Fever, productive cough, body ache

RECOMMENDED REGIMEN:

Symptom	Medication	Dosage	Notes
Fever/Body Ache	Acetaminophen	500mg q6-8h	Max 2g/day in CKD 3.
Productive Cough	Guaifenesin	200mg q4h	Safe.
	OR Acetylcysteine	600mg q12h	Safe. Mucolytic.
Nasal Congestion	Saline spray	As needed	Safe.
Sore Throat	Saline gargle	q4h	Safe.

MEDICATIONS TO AVOID:

X NSAIDs - ABSOLUTELY CONTRAINDICATED IN CKD

Even short courses (3-5 days) can cause significant AKI.

Triple Whammy Risk: NSAID + Diuretic + ACEi/ARB = High AKI risk.

X Pseudoephedrine - May reduce renal blood flow

X Phenylephrine - Same concern

FLUID MANAGEMENT:

- Maintain adequate hydration (critical in CKD)

- If patient on diuretic, monitor for dehydration with fever

- Do NOT fluid restrict unless specifically advised

SECTION 5: DRUG-DISEASE CONTRAINDICATIONS QUICK REFERENCE

5.1 ABSOLUTE CONTRAINDICATIONS

Drug/Drug Class	Condition	Risk
Pseudoephedrine	Uncontrolled HTN	Hypertensive crisis
Pseudoephedrine	Severe CAD	MI, arrhythmia
Pseudoephedrine	On MAOIs	Hypertensive crisis
Dextromethorphan	On MAOIs	Serotonin syndrome (fatal)
Dextromethorphan	On SSRIs/SNRIs	Serotonin syndrome
NSAIDs	CKD Stage 3-5	Acute kidney injury
NSAIDs	On anticoagulants	Major bleeding
NSAIDs	Peptic ulcer disease	GI bleeding
NSAIDs	Aspirin-sensitive asthma	Severe bronchospasm
Aspirin	Children with viral inf.	Reye's syndrome
Diphenhydramine	Angle-closure glaucoma	Acute glaucoma

5.2 RELATIVE CONTRAINDICATIONS (USE WITH EXTREME CAUTION)

Drug/Drug Class	Condition	Risk
Pseudoephedrine	Controlled HTN	BP elevation
Pseudoephedrine	On Triptans	Vasospasm
Pseudoephedrine	Anxiety disorders	Worsened anxiety
Pseudoephedrine	Hyperthyroidism	Tachycardia, HTN
Pseudoephedrine	BPH	Urinary retention
Dextromethorphan	On Bupropion	Increased DXM levels
Dextromethorphan	Liver disease	Accumulation
NSAIDs	HTN on ACEi/ARB	Reduced efficacy
NSAIDs	Heart failure	Fluid retention
NSAIDs	On diuretics	Reduced efficacy
Acetaminophen	Liver disease	Hepatotoxicity
Acetaminophen (>2g/day)	On Warfarin	Increased INR
First-gen Antihistamine	BPH	Urinary retention
First-gen Antihistamine	Elderly (>65)	Falls, confusion

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SECTION 6: PEDIATRIC & GERIATRIC CONSIDERATIONS

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6.1 GERIATRIC PATIENTS (>65 YEARS) - COLD TREATMENT

Special Considerations:

- Avoid first-generation antihistamines (Diphenhydramine): Increased fall risk, confusion, urinary retention, anticholinergic burden.
- Use second-generation: Loratadine, Cetirizine, Fexofenadine.
- Reduce doses: Start low, go slow.
- Acetaminophen: Max 2g/day in elderly (frailty, reduced hepatic function).
- Dextromethorphan: Safe but may cause confusion in very elderly.
- Avoid combination products: Multiple ingredients increase adverse effect risk.
- Pseudoephedrine: Use with extreme caution. Cardiovascular risk increased.

BEERS CRITERIA - Medications to AVOID in Elderly:

- Diphenhydramine (anticholinergic)
- NSAIDs (if CKD, heart failure, or GI risk)
- Pseudoephedrine (if cardiovascular disease)

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SECTION 7: CASE STUDIES & EXAMPLES

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CASE STUDY 1: COLD IN HYPERTENSION + DIABETES PATIENT

Patient: John, 58M, HTN on Lisinopril 10mg, DM2 on Metformin 500mg BID

Complaint: "I feel cold and sneezing"

Assessment: Viral URI. Mild symptoms. Vitals: BP 132/84, HR 78, Glucose 145.

RECOMMENDATION:

- Loratadine 10mg daily for sneezing/rhinorrhea
- Saline nasal spray PRN for congestion
- Acetaminophen 500mg q6h PRN if fever/body ache develops

- Guaifenesin 200mg q4h PRN if cough develops
- Continue Lisinopril and Metformin as prescribed
- AVOID: Pseudoephedrine, Phenylephrine, NSAIDs, combination cold products
- Monitor BP daily. If BP >150/95, contact physician.
- Monitor glucose q6h. Illness can increase glucose levels.
- Increase fluid intake (water, sugar-free). Rest.

RATIONALE:

- Loratadine: Non-drowsy antihistamine. No BP effect. No glucose effect.
- No interaction with Lisinopril or Metformin.
- Safe in both HTN and DM2.
- Avoiding pseudoephedrine prevents BP elevation.
- Acetaminophen chosen over NSAIDs for renal protection and BP control.

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CASE STUDY 2: COLD + FEVER IN ANTICOAGULATED PATIENT

Patient: James, 67M, AFib on Apixaban 5mg BID, HTN on Metoprolol 50mg, Gout
 Complaint: "Fever 101.2°F, body ache, sneezing, mild cough"
 Assessment: Viral syndrome. Vitals: BP 128/76, HR 82 irreg, Temp 101.2°F.

RECOMMENDATION:

- Acetaminophen 500mg q6h (MAX 2g/day) for fever and body ache
- Loratadine 10mg daily for sneezing
- Guaifenesin 200mg q4h PRN for cough
- Saline nasal spray PRN
- Continue Apixaban and Metoprolol as prescribed
- AVOID: NSAIDs (including Ibuprofen, Naproxen, Aspirin) - ABSOLUTELY CONTRAINDICATED
- AVOID: Pseudoephedrine - Opposes beta-blocker effects
- Monitor for bleeding signs (bruising, dark stools, nosebleeds, gum bleeding)
- If fever >102°F or persistent >3 days, seek medical attention.

RATIONALE:

- Acetaminophen safe with Apixaban at ≤2g/day.
- NSAIDs contraindicated due to synergistic bleeding risk.
- Acetaminophen does not affect uric acid (gout-safe).
- Pseudoephedrine avoided due to beta-blocker interaction and possible BP elevation.

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CASE STUDY 3: COUGH AND COLD IN PATIENT ON SSRI

Patient: Michael, 53M, HTN on Lisinopril 20mg, GERD on Omeprazole,
Anxiety/Depression on Sertraline 50mg
Complaint: "Cough (dry), runny nose, sneezing for 2 days"
Assessment: URI with predominant cough. Vitals: BP 130/82, HR 80.

RECOMMENDATION:

- Guaifenesin 200mg q4h for cough (NOT dextromethorphan!)
- Benzonatate 100mg TID PRN for persistent dry cough (alternative)
- Loratadine 10mg daily for sneezing and rhinorrhea
- Continue Lisinopril, Omeprazole, Sertraline
- AVOID: Dextromethorphan (SEROTONIN SYNDROME RISK with Sertraline)
- AVOID: Pseudoephedrine/Phenylephrine (BP elevation + possible anxiety)
- Monitor for serotonin syndrome symptoms (confusion, agitation, sweating, shivering)
- If persistent cough >7 days, re-evaluate for other causes.

RATIONALE:

- Dextromethorphan CONTRAINDICATED with Sertraline (both serotonin reuptake inhibitors)
- Guaifenesin safe. Benzonatate safe alternative.
- Loratadine safe with all current medications.
- No pseudoephedrine (HTN and potential anxiety exacerbation).

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CASE STUDY 4: COLD IN CKD + DIABETES PATIENT

Patient: Robert, 72M, CKD Stage 2, DM2 on Insulin Glargine, Hyperlipidemia
Complaint: "Fever 100.8°F, body ache, productive cough, thick phlegm"
Assessment: URI with bronchitis component. Labs: eGFR 58, Cr 1.4, HbA1c 7.1%.

RECOMMENDATION:

- Acetaminophen 650mg q8h (max 2g/day) for fever and body ache
- Guaifenesin 400mg q12h + Acetylcysteine 600mg q12h for productive cough

- Loratadine 10mg daily if sneezing present
- Increase fluid intake to maintain hydration
- Continue insulin, atorvastatin, furosemide
- AVOID: NSAIDs - NEPHROTOXIC in CKD
- AVOID: Dextromethorphan-containing products with high sugar
- Monitor blood glucose q4-6h (infection can cause hyperglycemia)
- Monitor urine output. If decreased, contact physician.
- If fever >102°F or breathing difficulty, seek immediate care.

RATIONALE:

- Acetaminophen is analgesic of choice in CKD.
- NSAIDs absolutely contraindicated due to nephrotoxicity risk in CKD.
- Guaifenesin and Acetylcysteine safe in CKD.
- Fluid management critical - maintain hydration.
- Triple Whammy Risk if NSAID used: NSAID + Furosemide + CKD = High AKI risk.

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CASE STUDY 5: COLD + ASTHMA PATIENT

Patient: Mary, 65F, Asthma on Albuterol PRN, HTN on Amlodipine, Allergies on Cetirizine

Complaint: "Sneezing, runny nose, itchy eyes - feels like allergies but different"

Assessment: Acute URI superimposed on allergic rhinitis. Vitals: BP 126/78, RR 16.

RECOMMENDATION:

- Continue Cetirizine 10mg daily (already on for allergies - can serve dual purpose)
- Add Oxymetazoline nasal spray BID (MAX 3 DAYS) for congestion
- Guaifenesin 200mg q4h PRN for post-nasal drip/cough
- Acetaminophen 500mg q6h PRN if fever
- Continue Amlodipine and Albuterol
- AVOID: Diphenhydramine (additive anticholinergic with possible asthma drying effect)
- AVOID: Aspirin (potential aspirin-sensitive asthma)
- AVOID: Pseudoephedrine (opposes amlodipine)
- Monitor peak flow daily during illness
- If wheezing increases, follow asthma action plan.

RATIONALE:

- Continue Cetirizine - no need for additional antihistamine.

- Oxymetazoline safe for short-term use (≤ 3 days) to avoid rhinitis medicamentosa.
- Aspirin avoided in asthmatics due to Samter's Triad potential.
- Pseudoephedrine can oppose calcium channel blocker effects.

SECTION 8: QUICK REFERENCE - SAFE MEDICATION BY COMORBIDITY

SAFE COLD MEDICATIONS BY CONDITION (Quick Lookup Table):

Condition	Analgesic	Antihistamine	Decongestant	Cough Med	Notes
HTN	Acetaminophen	Loratadine	Oxymetazoline	Guaifenesin	Avoid Pseudoephedrine
		Cetirizine	(nasal, max3d)	Dextrometh.	Avoid NSAIDs
Diabetes	Acetaminophen	Loratadine	Oxymetazoline	Guaifenesin	Avoid sugar syrups
		Cetirizine	(nasal)	Dextrometh.	Monitor glucose
CKD	Acetaminophen	Loratadine	Saline only	Guaifenesin	AVOID NSAIDs
	(max 2g/d)	(safe)		Dextrometh.	Adjust doses
AFib/Anticoag.	Acetaminophen	Loratadine	Saline only	Guaifenesin	AVOID NSAIDs
	(max 2g/d)	Cetirizine		Dextrometh.	Monitor bleeding
Asthma/COPD	Acetaminophen	Loratadine	Oxymetazoline	Guaifenesin	Continue inhalers
			(nasal, max3d)	Acetylcyst.	Avoid ASA if sens.
SSRI/SNRI	Acetaminophen	Loratadine	Oxymetazoline	Guaifenesin	AVOID Dextrometh.
		(nasal)		Benzonatate	Serotonin syndrome
Gout	Acetaminophen	Loratadine	Oxymetazoline	Guaifenesin	Avoid ASA high dose

			(nasal)	Dextrometh.	NSAIDs cautious	
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Hypothyroid		Acetaminophen	Loratadine	Oxymetazoline	Guaifenesin	All generally safe
			(nasal)	Dextrometh.		
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Migraine/Triptan		Acetaminophen	Loratadine	SALINE ONLY	Guaifenesin	AVOID Pseudoephedrine
				Dextrometh.	Vasospasm risk	
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GERD		Acetaminophen	Loratadine	Oxymetazoline	Guaifenesin	Avoid alcohol syrups
			(nasal)	Dextrometh.		
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Anxiety		Acetaminophen	Loratadine	Saline only	Guaifenesin	Avoid Pseudoephedrine
				Dextrometh.*	May worsen anxiety	
				(*if not SSRI)		

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END OF MEDICAL HANDBOOK
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